Jackson Runtian Ye

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Education

Rutgers, The State University of New Jersey New Brunswick, NJ

- Bachelor of Science in Electrical and Computer Engineering (Honors College) *May, 2021*
- Ph.D. in Electrical and Computer Engineering (Advised by Prof. Alexandre V. Morozov) *Expected Fall 2027*
  - Research interests: Generative AI Application, Remote Sensing, Symbolic Learning

Research Experience

Rutgers, The State University of New Jersey New Brunswick, NJ

Undergraduate Research Assistant with Professor Yingying Chen *Sep, 2021 - May, 2022*

- Participate in research and development in the areas of machine learning and signal processing
- Conduct experiments on voice assistant systems to investigate the relationship between the audio domain and the vibration domain
- Software development for experimental usages

Undergraduate Research Assistant with Professor Shiqing Ma *Jul, 2021 - Apr, 2022*

- Design and develop benchmarks to test Linux Kernel and software behaviors
- Conduct research on various system auditing techniques based on Transparent Computing Project by DARPA
- Build a specific audit policy and environment for testing purposes
- Develop and modify the Linux kernel module to improve Linux audit performance
- Customize eBPF and BCC application to retrieve system signals for Linux system behavior prediction powered by classic machine learning technique

University of Science and Technology of China Hefei, China

Research Assistant with Professor Fan Jiang *Summer, 2020*

- Conducted research and code analysis on data-network packet analyzer tcpdump
- Conducted detailed reverse engineering studies about malicious code and malware

Career

Rutgers, The State University of New Jersey New Brunswick, NJ

Teaching Assistant *Spring 2023*

- Able to effectively manage large groups of students. Committed to individual excellence and supporting each student in the way that helps them meet their learning objectives.

Creek Information and Technology *Shanghai, China*

Founder *May, 2020 - Jun, 2021*

- Provided strong leadership and effective interdepartmental communication to effectively meet or exceed all project timelines and budgetary goals
- Understood core business processes and identified technologies to support and enhance operations

Huami *Hefei, China*

Information Technology Internship *May, 2017– Aug, 2017*

- Assisted end-users with hardware and software configuration, setup, and troubleshooting
- Performed Level 1 and minor Level 2 technical support consistently resolving users' issues
- Conducted technology inventory management to maintain workflow

FLYTEK Co., Ltd *Hefei, China*

President's Office Assistant Internship *Summer 2015*

- Worked under the Office of the CTO to develop organizational systems for contracts, records, reports, and agendas.
- Drafted meeting agendas, supplied advance materials, and executed follow-up for meetings and team conferences.
- Verified operation of office equipment by completing preventive maintenance requirements and calling for repairs.

Merits

- Rutgers SAS Excellence Fellowship
- Columbia CS3 Streetscape Validate Accelerator Programe
- Eligible for NSF Foundation I-Corp Grant (Upon Program Completion)

Projects

- Population Estimation with Object Centric Vision Transformer State Space Model
- Physical Law Inference with Symbolic Learning
- Built a Kubernetes pen-testing lab with Docker for Graduation Capstone
- Finished 41 machines in the PWK (Offensive Security) Lab
- Translate Metasploit Ruby code to stand-alone Python Scripts
- Constantly practice on HackTheBox platform

Language & Skills

- Skilled with Metasploit Framework and Virtualization Environment
- Writing scripts and tools to aid in the penetration testing process, cross-compiling and modifying public exploit code
- Familiar with route network traffic systems: OpenWRT, and RouterOS
- C++, Python, Shell Script, Java, MatLab, LaTex, HTML, CSS